

Machine Learning & Python

Lecture Notes

Lecture 13 — Correlation Analysis (04/09/2006)

Topics Covered

- What is Correlation Analysis? (What, Why, How)
- Interpreting Correlation Values
- Uses of Correlation Analysis
- Data Visualization

1. What is Correlation Analysis?

Correlation analysis is used to determine the **relationship between two or more variables** in a dataset. It helps us understand how changes in one variable affect another variable.

Important Note: Correlation is used between **two columns**. IQR, Mean, and Median are used on a **single column**.

2. Interpreting Correlation Values

- Correlation values close to **+1** or **-1** indicate a **strong** relationship.
- Correlation values close to **0** indicate a **weak or no** relationship.
- The **magnitude** represents the strength; the **sign** indicates direction.
 - **+0.8 to +1.0** — Strong Positive
 - **+0.4 to +0.7** — Moderate Positive
 - **-0.1 to +0.1** — Weak / No relationship
 - **-0.4 to -0.7** — Moderate Negative
 - **-0.8 to -1.0** — Strong Negative

3. Uses of Correlation Analysis

- **Predictive Modeling** — If two variables are correlated, one can predict the other.
- **Feature Selection** — Drop redundant (highly correlated) features to simplify ML models.
- **Data Exploration** — Quickly discover which variables are related before model building.

Python Code — Correlation Analysis

Step 1: Create a DataFrame & calculate correlation

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

data = {
    'Temperature': [30, 32, 28, 35, 33, 25, 27, 31],
    'Ice_Cream_Sales': [200, 220, 180, 260, 240, 150, 170, 210],
    'Umbrella_Sales': [10, 8, 20, 5, 6, 30, 25, 12],
}
df = pd.DataFrame(data)

# Single pair
r = df['Temperature'].corr(df['Ice_Cream_Sales'])
print(f"r = {r:.2f}")          # Output: r = 0.99

# Full matrix
print(df.corr())
```

Step 2: Heatmap & Scatter Plot

```
# Heatmap
sns.heatmap(df.corr(), annot=True, fmt='.2f',
            cmap='coolwarm', vmin=-1, vmax=1)
plt.title("Correlation Heatmap")
plt.show()

# Scatter plot
plt.scatter(df['Temperature'], df['Ice_Cream_Sales'], color='teal')
plt.xlabel("Temperature"); plt.ylabel("Ice Cream Sales")
plt.title(f"Scatter Plot (r = {r:.2f})")
plt.show()
```

Data Visualization

Data visualization is the **graphical representation** of data and information. It involves creating visual elements such as charts, graphs, and maps to help people understand complex data patterns.

Common Chart Types

- **Line Chart** — Trends over time — `plt.plot()`
- **Bar Chart** — Comparing categories — `sns.barplot()`
- **Scatter Plot** — Relationship between 2 variables — `plt.scatter()`
- **Histogram** — Distribution of one variable — `sns.histplot()`
- **Box Plot** — Median, IQR, outliers — `sns.boxplot()`
- **Heatmap** — Correlation matrix — `sns.heatmap()`

Quick Visualization Code

```
tips = sns.load_dataset("tips")           # built-in sample dataset

# 1. Histogram
sns.histplot(tips['total_bill'], kde=True)
plt.title("Distribution of Total Bill"); plt.show()

# 2. Box Plot
sns.boxplot(x='day', y='total_bill', data=tips)
plt.title("Total Bill by Day"); plt.show()

# 3. Full correlation heatmap
sns.heatmap(tips.select_dtypes('number').corr(),
            annot=True, cmap='coolwarm')
plt.title("Heatmap"); plt.show()
```

Quick Revision — Key Points

- r ranges from **-1** to **+1**
- $|r| > 0.7$ = Strong | $0.4-0.7$ = Moderate | < 0.4 = Weak
- Positive $r \rightarrow$ both variables increase together
- Negative $r \rightarrow$ one increases, other decreases
- **Correlation $\neq$ Causation**
- Use **df.corr()** for the matrix; **sns.heatmap()** to visualise
- Correlation: 2 columns | Mean/Median/IQR: 1 column